Hyperparameter Sensitivity Trends Across Datasets in the UCI Machine Learning Repository for Classification Tasks

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ABSTRACT

Machine learning model performance depends heavily on hyperparameters, yet benchmark studies typically report only the best-performing settings rather than the sensitivity of models to hyperparameter variation. Hyperparameter sensitivity is critical to understand because it determines the tuning effort required and the robustness of predictive performance. This study evaluates hyperparameter sensitivity across five UCI ML Repository datasets (Iris, Wine Quality, Breast Cancer Wisconsin (Diagnostic), Heart Disease - Cleveland, and Adult) using five classifiers: Logistic Regression, k-Nearest Neighbours (KNN), Decision Trees, Support Vector Machines (SVM), and Multi-Layer Perceptrons (MLP). Sensitivity is quantified using the variance of Accuracy, F1-score, and AUC across hyperparameter configurations. The experimental results show that Logistic Regression and KNN remain relatively stable under hyperparameter variations, while SVM and MLP demonstrate significant performance variability across hyperparameter settings, especially on Heart Disease and Breast Cancer datasets. Dataset characteristics, such as dimensionality, noise, and class imbalance, strongly influence sensitivity trends. These findings guide practitioners in prioritizing hyperparameters that require tuning and choosing models that remain reliable under limited tuning budgets.

1. INTRODUCTION

Hyperparameters play a crucial role in determining the performance, stability, and generalization capability of machine learning models. Unlike parameters learned from data, hyperparameters must be chosen manually or via cross validation, and their stability heavily influences the reliability of the model. Although common ML benchmarks compare models using optimal hyperparameter settings, they overlook an essential real-world concern, which is the variance in performance with changing hyperparameter values.

Understanding hyperparameter sensitivity helps to:

- determine tuning effort required for different hyperparameters
- anticipate model robustness
- identify datasets that require careful hyperparameter settings
- avoid overfitting by recognizing unstable models
- select models aligned with complexity and noise levels of the data

This project systematically evaluates hyperparameter sensitivity across diverse UCI ML Repository datasets and classifiers, using variance across performance metrics to quantify stability. Results reveal strong relationships between dataset properties, model type, and sensitivity patterns.

2. RELATED WORK

Several studies have examined hyperparameter optimization strategies, including grid search, random search, and Bayesian optimization.

Moshe Sipper (2022) conducted a large-scale investigation involving 26 ML algorithms, 250 datasets (covering regression, binary, and multinomial classification), 6 performance metrics, and 28,857,600 algorithm runs to assess the role of hyperparameter tuning in the ML pipeline. In his paper, *"High Per Parameter: A Large-Scale Study of Hyperparameter Tuning for Machine Learning Algorithms"*, he concludes that while significant gains from tuning are not always guaranteed, default hyperparameter settings can perform poorly for certain datasets and algorithms. Finally, the study ranks the 26 algorithms by their responsiveness to tuning, with Random Forest and Decision Tree among the most sensitive, while SVM, Logistic Regression, and KNN appear less affected.

Probst et al. (2019), in their paper *"Tunability: Importance of Hyperparameters of Machine Learning Algorithms"* conducted a similar study involving 6 common classifiers, 38 datasets from the OpenML platform and 2 performance metrics - accuracy and AUC. They introduced the concept of tunability for hyperparameters and classifiers, which is how much the performance can be improved by tuning hyperparameters of a model, relative to using default values. Their results found that SVM and XGBoost were among the most tunable models, whereas KNN and Random Forest were relatively less tunable.

Bergstra & Bengio (2012) demonstrated that randomly chosen trials are more efficient for hyper-parameter optimization than trials on a grid. Their key insight was that not all hyperparameters equally influence model performance, and their importance varies across datasets, highlighting the asymmetric sensitivity of hyperparameters.

While these studies address the value and strategies of hyperparameter tuning, few directly analyze sensitivity, that is, the degree of metric variability resulting from changes in individual hyperparameters, across multiple classifiers and diverse datasets. Our work specifically focuses on this gap by empirically quantifying hyperparameter sensitivity for five popular classification models across five well-known UCI datasets, using accuracy, AUC, and F1 score as evaluation metrics.

3. METHODOLOGY

3.1 Datasets

We used five datasets from the UCI Machine Learning Repository:

Dataset	**Samples**	**Task and Target**	**Features**	**Class Balance**	**Other Characteristics**
Iris	150	Multi-class classification (Sentosa, Versicolor, Virginica)	4 numeric	Balanced (50 each)	Small, Clean, No missing values, Low Noise, Simple structure
Heart Disease	303	Binary classification (Disease / No Disease)	13 numerical and categorical	Slightly imbalanced (0: 164, 1: 139)	Small, Mixed feature types, Very few missing values, Moderate noise
Wine Quality	Red: 1,599 White: 4,898	Multi-class classification (Numerical quality scores converted to class labels: low = (quality<5), med = (quality=6/7), high = (quality>7)	11 numerical, 1 categorical	Highly imbalanced (most samples cluster around low- and mid-quality) (low: 2384, med: 3915, high: 198)	Medium-sized, Clean, Continuous features, No missing values, Noisy
Breast Cancer Wisconsin (Diagnostic)	569	Binary classification (Malignant / Benign)	30 numerical	Imbalanced (approx. 37% malignant, 63% benign)	Medium-size, Smooth numeric features, High separation between classes

Adult	48,842	Binary classification (>50K / <= 50K)	14 numerical and categorical	Imbalanced (~24% >50K)	Large, High - dimensional categorical data, Noisy, Missing values

3.2 Preprocessing

- **Missing values** were checked and handled by imputing or removing the row
- **Categorical encoding** was used (One-hot Encoding for features and Label Encoding for target class)
- **Standardization** was applied for numerical features
- **Data was split** into features and target, however, train-test split was not done as it is unnecessary for the project's purpose
- **5-fold or 10-fold cross-validation** ensured stable evaluation

3.3 Classifiers & Hyperparameters

Logistic Regression

- Penalty: none, l1, l2, elasticnet
- Solver: liblinear, lbfgs, saga
- Regularization Strength : 0.001, 0.01, 0.1, 1, 10, 100, 1000
- L1 Ratio: 0.1, 0.2 ,0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 (elasticnet)

KNN

- k: 1 → 31
- Weight Metrics: uniform, distance
- Algorithms: auto, ball_tree, kd_tree, brute
- Distance Metrics: Euclidean, Manhattan, Minkowski, Chebyshev, Cosine

Decision Tree

- Criteria: gini, entropy, log loss
- Max depth: None, 2, 3, 5, 10, 15, 20
- Min samples split: 1, 2, 5, 8, 10

- Min samples leaf: 1, 2, 5, 8, 10
- Max features: None, sqrt, log2

SVM

- C: 0.001, 0.01, 0.1, 1, 10, 100, 1000
- Kernel: linear, polynomial, RBF, sigmoid
- Gamma: scale, auto, 0.001, 0.01, 0.1, 1
- Degree (for polynomial kernel): 2, 3, 4, 5

MLP

- Hidden layers: (10,), (50,), (100,), (10,10), (50,50), (100,50)
- Activation: identity, logistic, tanh, relu
- Alpha: 0.001, 0.01, 0.1, 1, 10, 100, 1000
- Learning Rate: constant, invscaling, adaptive (for sgd solver only)
- Initial Learning Rate: 0.0001, 0.001, 0.01, 0.1

3.4 Sensitivity Computation

For each hyperparameter (h), we compute sensitivity as:

$$\text{Sensitivity}(h) = \text{Var}(M(h_i))$$

where (M) is Accuracy, AUC, or F1 Score, and (h_i) are sampled hyperparameter values.

Variance was selected as the sensitivity metric because it directly captures dispersion of performance under controlled hyperparameter variations. Low variance indicates robustness under tuning uncertainty, while high variance suggests unstable generalization behavior. Variance-based sensitivity is model-agnostic and allows fair comparison across diverse classifiers and datasets.

4. EXPERIMENTAL RESULTS AND ANALYSIS

Complete tables listing all hyperparameter combinations along with their corresponding Accuracy, AUC, and F1 Score values are available in our Python notebooks. Below, we present summarised tables

showing the sensitivity of each hyperparameter, calculated as the variance of the respective evaluation metric across its tested values.

4.1 Logistic Regression Sensitivity Summary

Accuracy Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Solver	0.0009481481	0.0000037037	0.0000000317	0.0001033298	0.0000000085	**0.0002110444**
Penalty	0.0001000000	0.0000296325	0.0000004499	0.0001411436	0.0000000075	**0.0000542467**
C	0.0038476190	0.0025910449	0.0000327925	0.0000140889	0.0000049868	**0.0012981064**
L1 Ratio	0.0000049383	0.0000160494	0.0000000204	0.0000000000	0.0000000012	**0.0000042018**
Average per Dataset	**0.0012251764**	**0.0006601076**	**0.0000083236**	**0.0000646406**	**0.0000012510**	

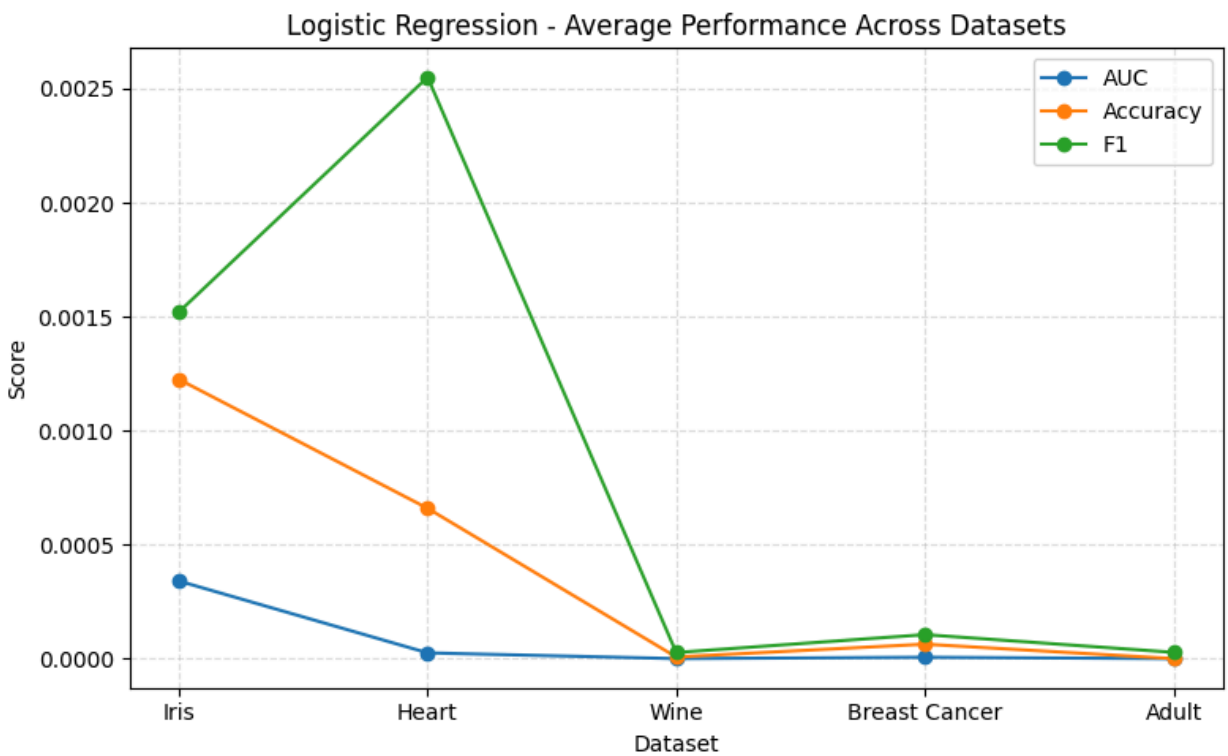
AUC Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Solver	0.0002028148	0.0000012614	0.0000001586	0.0000115325	0.0000000005	**0.0000431536**
Penalty	0.0000004444	0.0000097207	0.0000000631	0.0000133377	0.0000000002	**0.0000047132**
C	0.0011618413	0.0000940362	0.0000046203	0.0000036634	0.0000043958	**0.0002537114**
L1 Ratio	0.0000000000	0.0000007324	0.0000000033	0.0000000000	0.0000000001	**0.0000001472**
Average per Dataset	**0.0003412751**	**0.0000264377**	**0.0000012113**	**0.0000071334**	**0.0000010991**	

F1 Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Solver	0.0010378640	0.0000071006	0.0000000465	0.0001679706	0.0000000416	**0.0002426047**
Penalty	0.0001033953	0.0000309031	0.0000004444	0.0002312948	0.0000000343	**0.0000732144**
C	0.0049382057	0.0101413254	0.0001114937	0.0000264480	0.0001150695	**0.0030665085**
L1 Ratio	0.0000052421	0.0000197349	0.0000000166	0.0000000000	0.0000000043	**0.0000049996**
Average per Dataset	**0.0015211768**	**0.0025497660**	**0.0000280003**	**0.0001064284**	**0.0000287874**	

- **Most sensitive hyperparameter:** Regularization Strength
- **Least sensitive hyperparameter:** Penalty and L1 ratio
- **Most sensitive dataset:** Iris and Heart Disease
- **Most stable dataset:** Wine Quality and Adult



4.2 KNN Sensitivity Summary

Accuracy Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
K	0.0000544803	0.0004567201	0.0006243518	0.0001295960	0.0000814021	**0.0002693100**
Weight Metric	0.0000000000	0.0000000000	0.0000189539	0.0000015389	0.0000157357	**0.0000072457**
Algorithm	0.0000000000	0.0000000000	0.0000000000	0.0000000000	0.0000000000	**0.0000000000**
Distance Metric	0.0016311111	0.0028978287	0.0000103302	0.0030070421	0.0000215182	**0.0015135660**
Average per Dataset	**0.0004213978**	**0.0008386372**	**0.0001634090**	**0.0007845442**	**0.0000296640**	

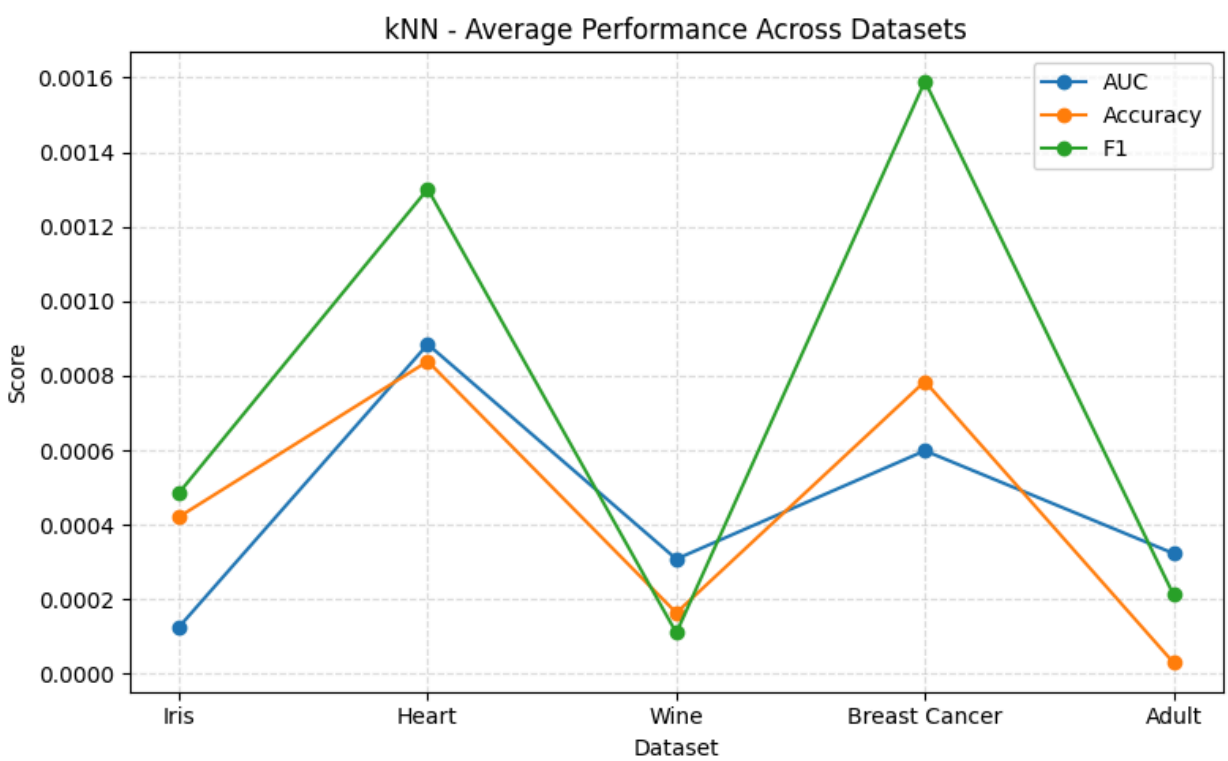
AUC Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
K	0.0000642124	0.0011268706	0.0011947437	0.0004156530	0.0012141772	**0.0008031314**
Weight Metric	0.0000008889	0.0000010577	0.0000067041	0.0000767695	0.0000010617	**0.0000172964**
Algorithm	0.0000000000	0.0000000000	0.0000000000	0.0000000000	0.0000000000	**0.0000000000**
Distance Metric	0.0004382778	0.0024094956	0.0000296216	0.0019020698	0.0000766773	**0.0009712284**
Average per Dataset	**0.0001258448**	**0.0008843560**	**0.0003077674**	**0.0005986231**	**0.0003229791**	

F1 Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
K	0.0000586871	0.0008123209	0.0004143203	0.0004524083	0.0007016203	**0.0004878714**
Weight Metric	0.0000000000	0.0000000000	0.0000093976	0.0000038509	0.0000416594	**0.0000109816**
Algorithm	0.0000000000	0.0000000000	0.0000000000	0.0000000000	0.0000000000	**0.0000000000**
Distance Metric	0.0018850569	0.0043923337	0.0000151230	0.0059057448	0.0001050281	**0.0024606573**
Average per Dataset	**0.0004859360**	**0.0013011637**	**0.0001097102**	**0.0015905010**	**0.0002120769**	

- **Most sensitive hyperparameter:** Distance Metric and K
- **Least sensitive hyperparameter:** Algorithm and Weight Metric
- **Most sensitive dataset:** Heart Disease and Breast Cancer
- **Most stable dataset:** Wine Quality and Adult



4.3 Decision Tree Sensitivity Summary

Accuracy Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Criteria	0.0000000000	0.0000508267	0.0000124054	0.0001025957	0.0000020454	**0.0000335746**
Max Depth	0.0000148148	0.0003462129	0.0020556150	0.0000165091	0.0002318430	**0.0005329990**
Min Samples Split	0.0000088889	0.0006066042	0.0000370644	0.0000493005	0.0000400846	**0.0001483885**
Min Samples Leaf	0.0000133333	0.0005748156	0.0002085162	0.0001810376	0.0001262244	**0.0002207854**
Max Features	0.0001333333	0.0002779165	0.0000770287	0.0000941536	0.0000231196	**0.0001211104**
Average per Dataset	**0.0000340741**	**0.0003712752**	**0.0004781259**	**0.0000887193**	**0.0000846634**	

AUC Variance Across Hyperparameters

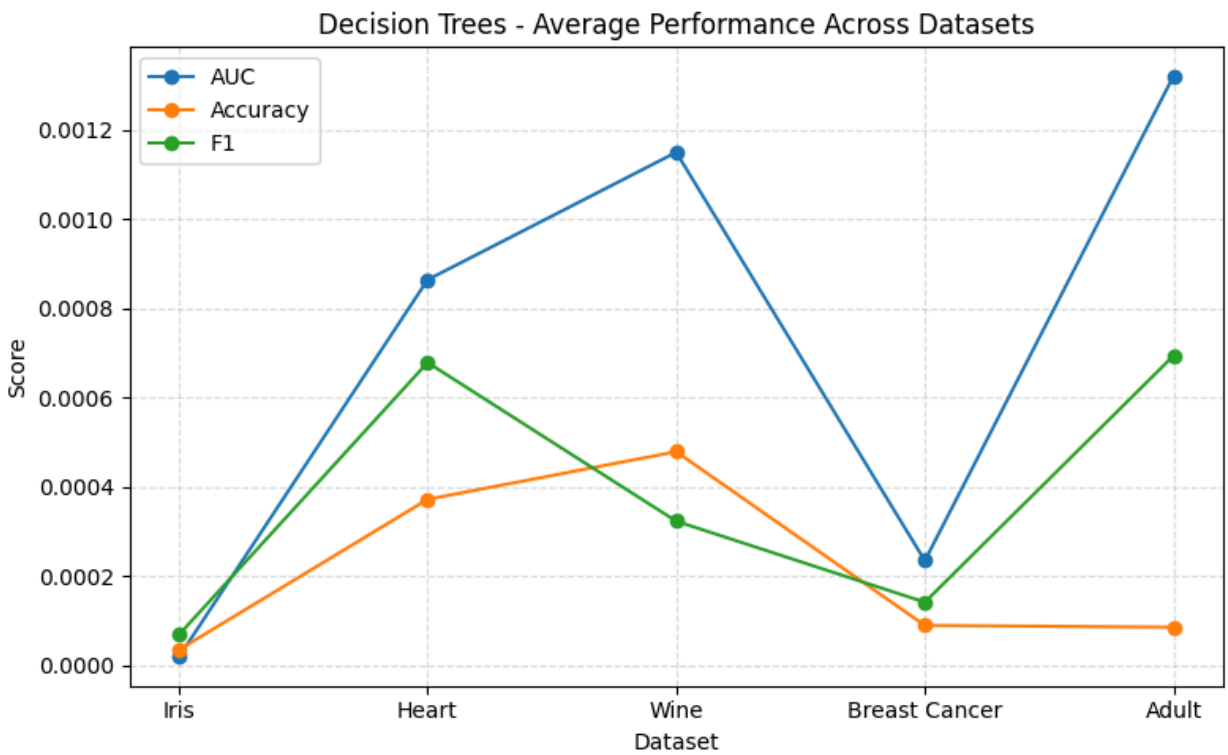
Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Criteria	0.0000000000	0.0003159462	0.0000304155	0.0001359790	0.0000040731	**0.0000972828**
Max Depth	0.0000282381	0.0011013138	0.0047748306	0.0000662092	0.0024872137	**0.0016915611**
Min Samples Split	0.0000304222	0.0009677749	0.0002769513	0.0001380225	0.0011880414	**0.0005202425**

Min Samples Leaf	0.0000248000	0.0016706891	0.0005727063	0.0007787228	0.0028468345	**0.0011787506**
Max Features	0.0000083333	0.0002615908	0.0000908274	0.0000502223	0.0000737709	**0.0000969489**
Average per Dataset	**0.0000183587**	**0.0008634630**	**0.0011491462**	**0.0002338312**	**0.0013199867**	

F1 Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Criteria	0.0000000000	0.0000458143	0.0000070674	0.0002040011	0.0000089540	**0.0000531674**
Max Depth	0.0000157832	0.0010648406	0.0013793624	0.0000337774	0.0030503690	**0.0011088265**
Min Samples Split	0.0000087102	0.0004735691	0.0000083501	0.0000770032	0.0000563369	**0.0001247939**
Min Samples Leaf	0.0000227266	0.0003644428	0.0002084999	0.0002720873	0.0002008824	**0.0002137278**
Max Features	0.0002923633	0.0014410670	0.0000082135	0.0001169804	0.0001486436	**0.0004014535**
Average per Dataset	**0.0000679167**	**0.0006779468**	**0.0003222987**	**0.0001407699**	**0.0006930372**	

- **Most sensitive hyperparameter:** Max Depth
- **Least sensitive hyperparameter:** Criteria and Max Features
- **Most sensitive dataset:** Wine Quality and Adult
- **Most stable dataset:** Iris



4.4 SVM Sensitivity Summary

Accuracy Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
C	0.0008063492	0.0158391353	0.0007901590	0.0158152803		**0.0083127309**
Kernel	0.0008111111	0.0008385247	0.0009947083	0.0153452814	0.0001700477	**0.0036319346**
Degree	0.0026962963	0.0009847487	0.0001280257	0.0005027952	0.0003355840	**0.0009294900**
Gamma	0.0023422222	0.0061017262	0.0000836109	0.0114346744	0.0000001043	**0.0039924676**
Average per Dataset	**0.0016639947**	**0.0059410337**	**0.0004991260**	**0.0107745078**	**0.0001685786**	

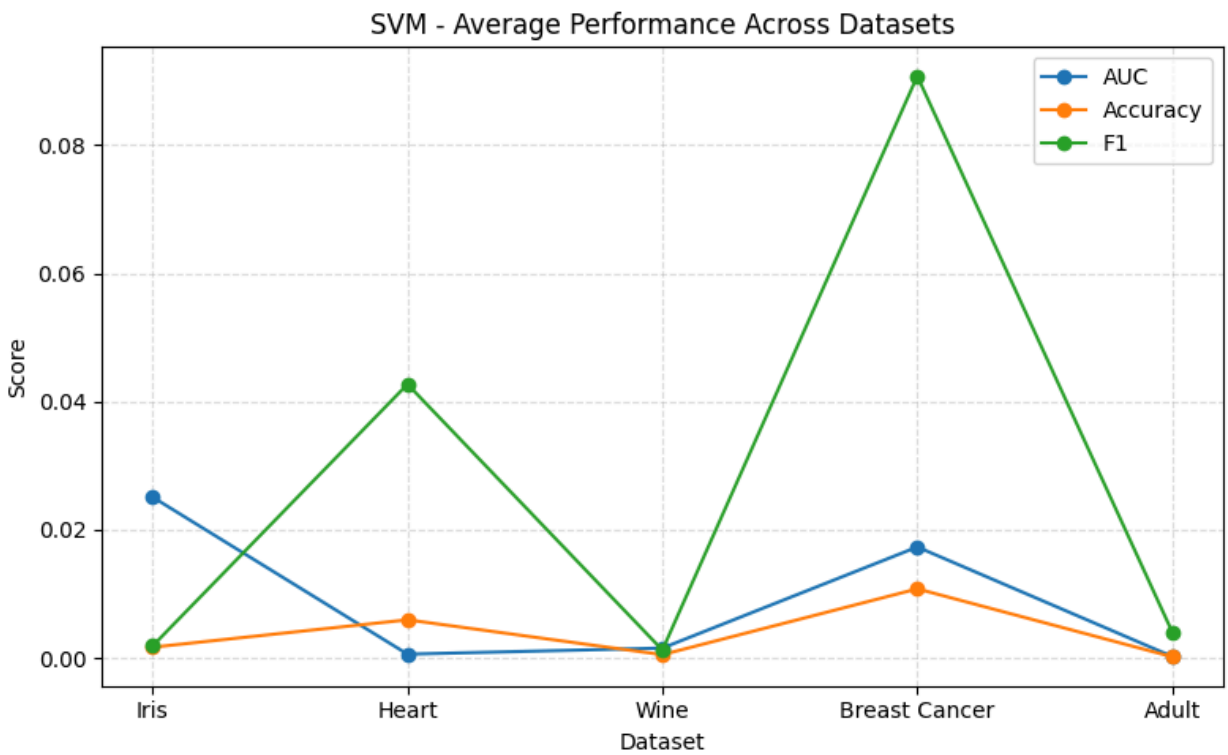
AUC Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
C	0.0936424974	0.0005206989	0.0018235693	0.0010884055		**0.0242687928**
Kernel	0.0001067778	0.0001654696	0.0029714373	0.0105142981	0.0004295924	**0.0028375150**
Degree	0.0005441111	0.0002588796	0.0001548088	0.0165634425	0.0002191810	**0.0035480846**
Gamma	0.0060507852	0.0013788047	0.0011574171	0.0410522794	0.0000189716	**0.0099316515**
Average per Dataset	**0.0250860429**	**0.0005809632**	**0.0015268081**	**0.0173046064**	**0.0002225816**	

F1 Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
C	0.0008280505	0.1423842539	0.0037553554	0.1629976738		**0.0774913334**
Kernel	0.0009749321	0.0018440471	0.0006561505	0.0736339530	0.0009222520	**0.0156062669**
Degree	0.0032311538	0.0028270457	0.0004127602	0.0077652179	0.0112101541	**0.0050892663**
Gamma	0.0024323285	0.0237259213	0.0000729811	0.1186785137	0.0000150049	**0.0289849499**
Average per Dataset	**0.0018666162**	**0.0426953170**	**0.0012243118**	**0.0907688396**	**0.0040491370**	

- **Most sensitive hyperparameter:** C and Gamma
- **Least sensitive hyperparameter:** Kernel
- **Most sensitive dataset:** Breast Cancer and Heart
- **Most stable dataset:** Wine Quality and Adult



4.5 MLP Sensitivity Summary

Accuracy Variance Across Hyperparameters

Hyperparameter	Iris	Heart Disease	Wine Quality	Breast Cancer	Adult	Average per HP
Hidden Layer Sizes	0.0001200000	0.0002933357	0.0010037201	0.0000403153	0.0001808254	**0.0003276393**
Activation	0.0000407407	0.0001401363	0.0002625499	0.0013114743	0.0002063716	**0.0003922546**
Alpha	0.0520373016	0.0165155672	0.0003293342	0.0008019685	0.0014689429	**0.0142306229**
Learning Rate	0.1083111111	0.0277725263	0.0002649095	0.0065925312	0.0003479638	**0.0286578084**
Learning Rate Init	0.0000111111	0.0004049658	0.0004638651	0.0097366873	0.0000376361	**0.0021308531**

Average per Dataset	**0.0321040529**	**0.0090253063**	**0.0004648758**	**0.0036965953**	**0.0004483480**	

AUC Variance Across Hyperparameters

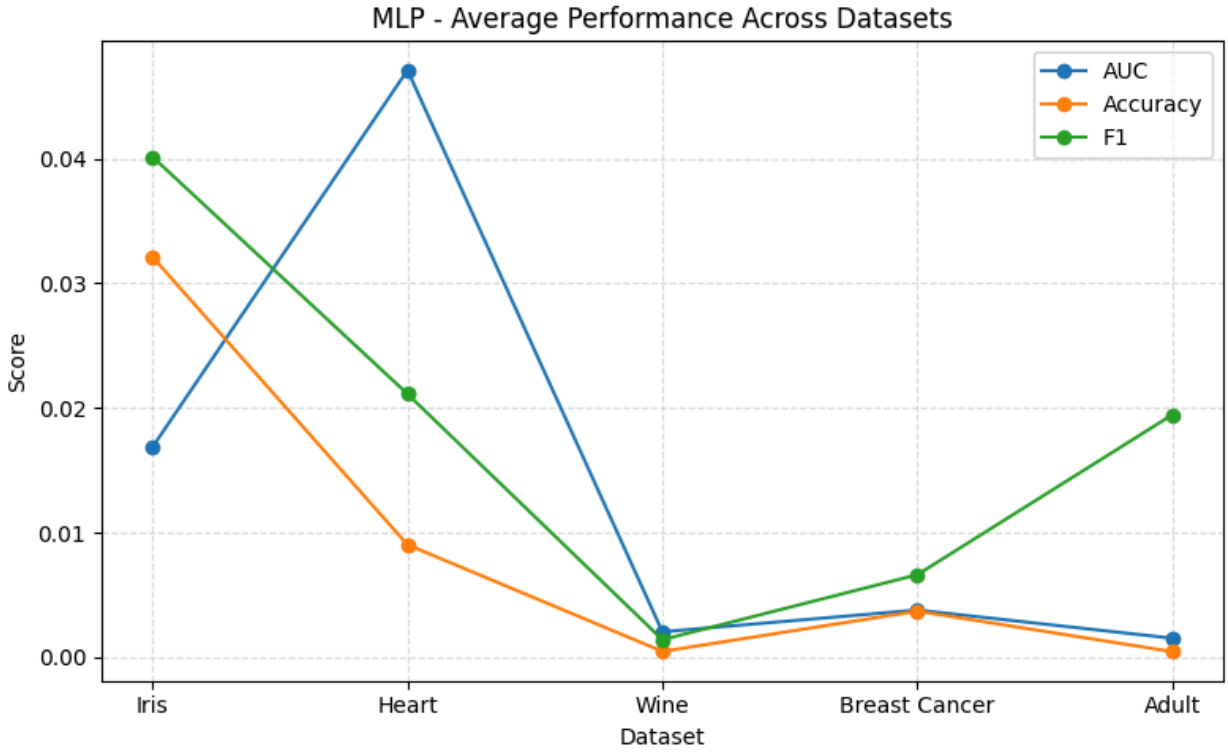
Hyperparameter	**Iris**	**Heart Disease**	**Wine Quality**	**Breast Cancer**	**Adult**	**Average per HP**
Hidden Layer Sizes	0.0000185222	0.0003603326	0.0019912398	0.0000220765	0.0003349217	**0.0005454186**
Activation	0.0000000000	0.0001947085	0.0007427578	0.0005682166	0.0003403085	**0.0003691983**
Alpha	0.0041671111	0.1331740431	0.0006907270	0.0008234815	0.0057089579	**0.0289128641**
Learning Rate	0.0801423704	0.1007660816	0.0065339340	0.0055698368	0.0011635686	**0.0388351583**
Learning Rate Init	0.0000080000	0.0009663519	0.0002493920	0.0119435507	0.0000816273	**0.0026497844**
Average per Dataset	**0.0168672007**	**0.0470923035**	**0.0020416101**	**0.0037854324**	**0.0015258768**	

F1 Variance Across Hyperparameters

Hyperparameter	**Iris**	**Heart Disease**	**Wine Quality**	**Breast Cancer**	**Adult**	**Average**
Hidden Layer Sizes	0.0000591532	0.0002218567	0.0005438877	0.0000708274	0.0005257376	**0.0002842925**
Activation	0.0000148892	0.0004601562	0.0002353355	0.0027606181	0.0002538213	**0.0007449640**
Alpha	0.0827483362	0.0004068938	0.0051238946	0.0018494974	0.0894822259	**0.0359221696**
Learning Rate	0.1177438575	0.1040825287	0.0007823562	0.0118311778	0.0068943867	**0.0482668614**

Learning Rate Init	0.0001340025	0.0004314030	0.0003466983	0.0165871915	0.0000783562	**0.0035155303**
Average per Dataset	**0.0401400477**	**0.0211205677**	**0.0014064344**	**0.0066198624**	**0.0194469056**	

- **Most sensitive hyperparameter:** Learning Rate
- **Least sensitive hyperparameter:** Hidden Layers Size and Activation
- **Most sensitive dataset:** Iris and Heart Disease
- **Most stable dataset:** Wine Quality and Adult



4.6 Cross-Model Sensitivity Insights

Most Sensitive Models:

1. MLP (Very sensitive to hyperparameters like alpha and learning rate due to complex architecture)
2. SVM (Sensitive especially to C and gamma; performance varies widely)

Most Stable Models:

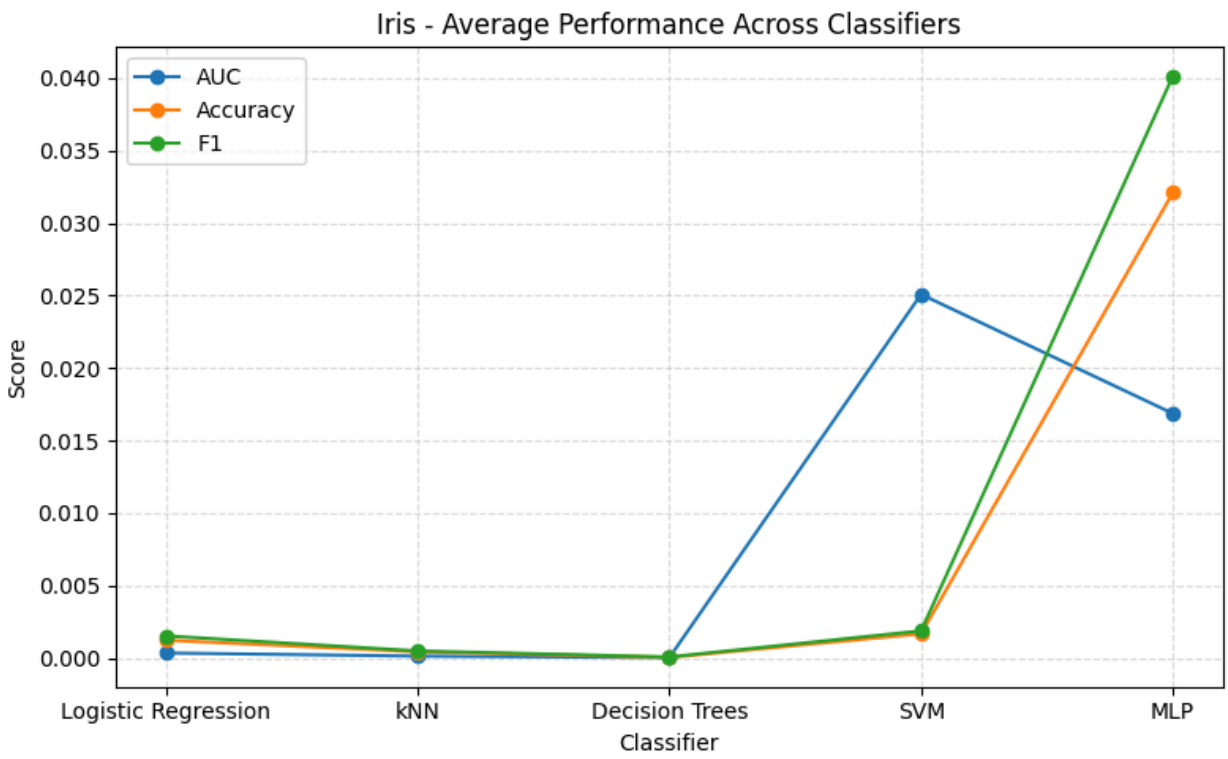
1. Logistic Regression (Simpler model with linear boundaries; fewer impactful hyperparameters)
2. KNN (Stability from consistent performance across distance/weight settings)
3. Decision Tree (Only sensitive to Max Depth, otherwise very stable)

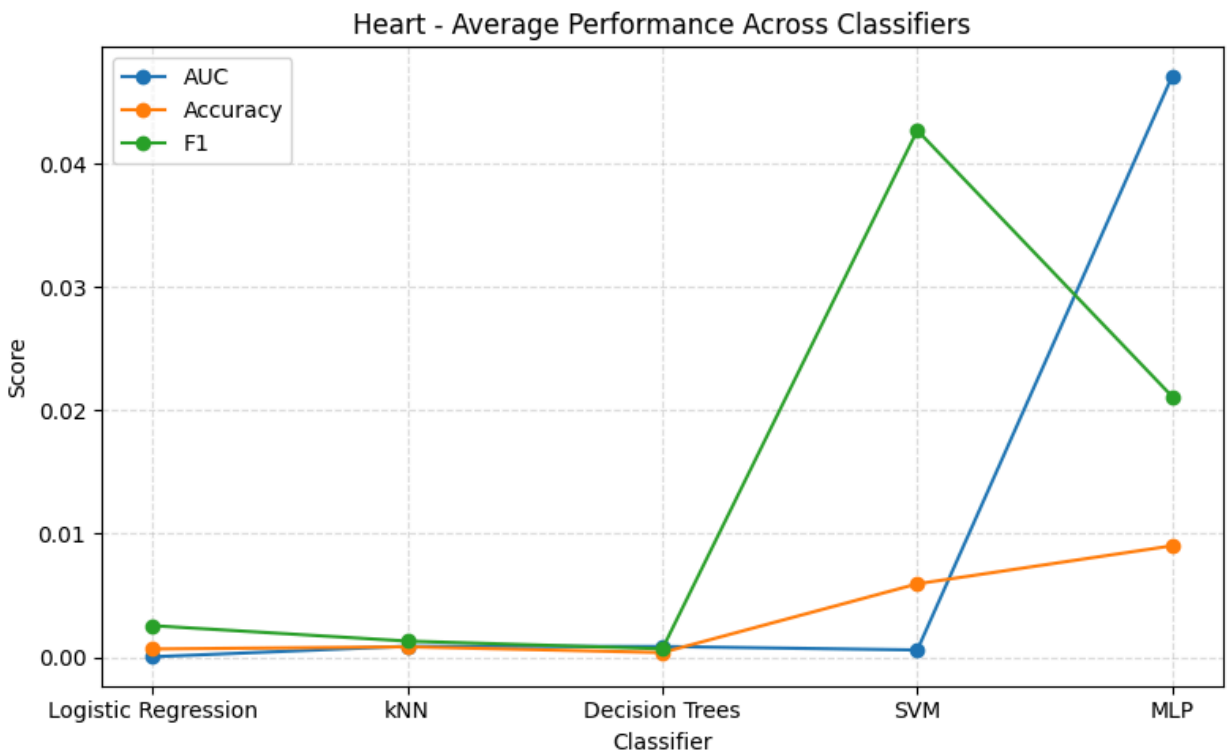
Most Sensitive Datasets:

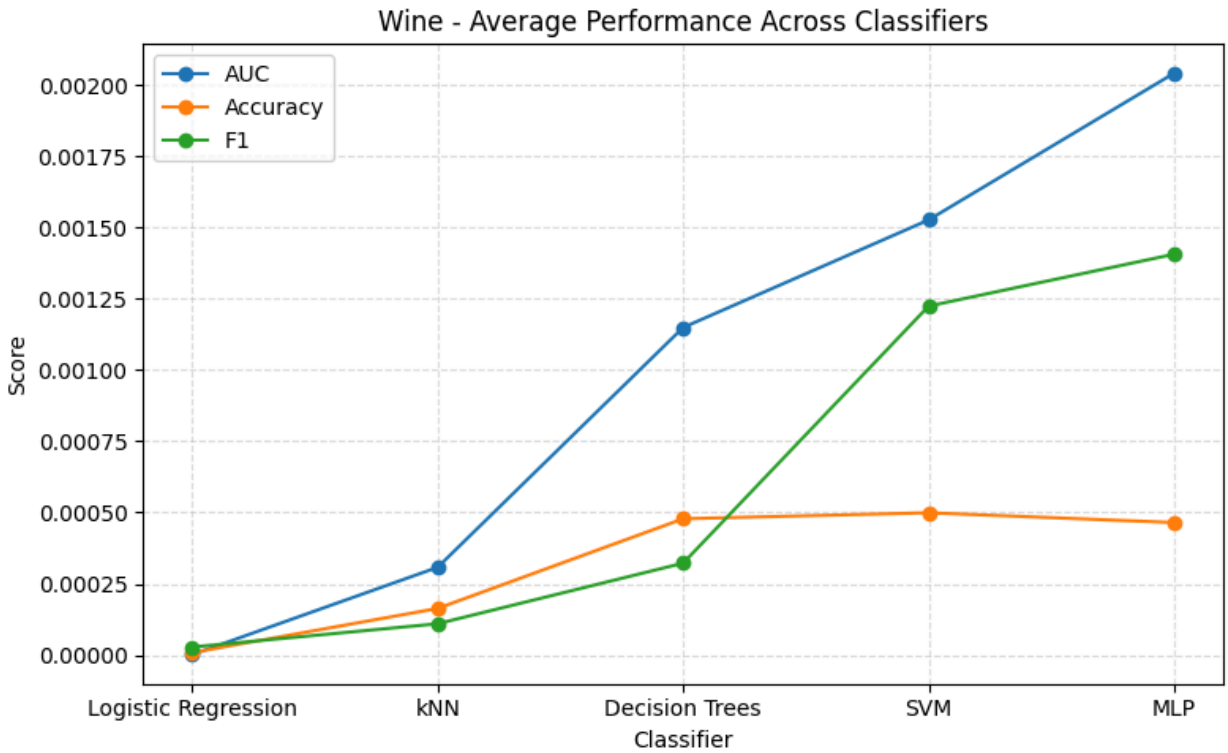
1. Heart Disease (Small, imbalanced dataset with non-linear relationships)
2. Breast Cancer (Sharp class boundaries increase variance with certain models)

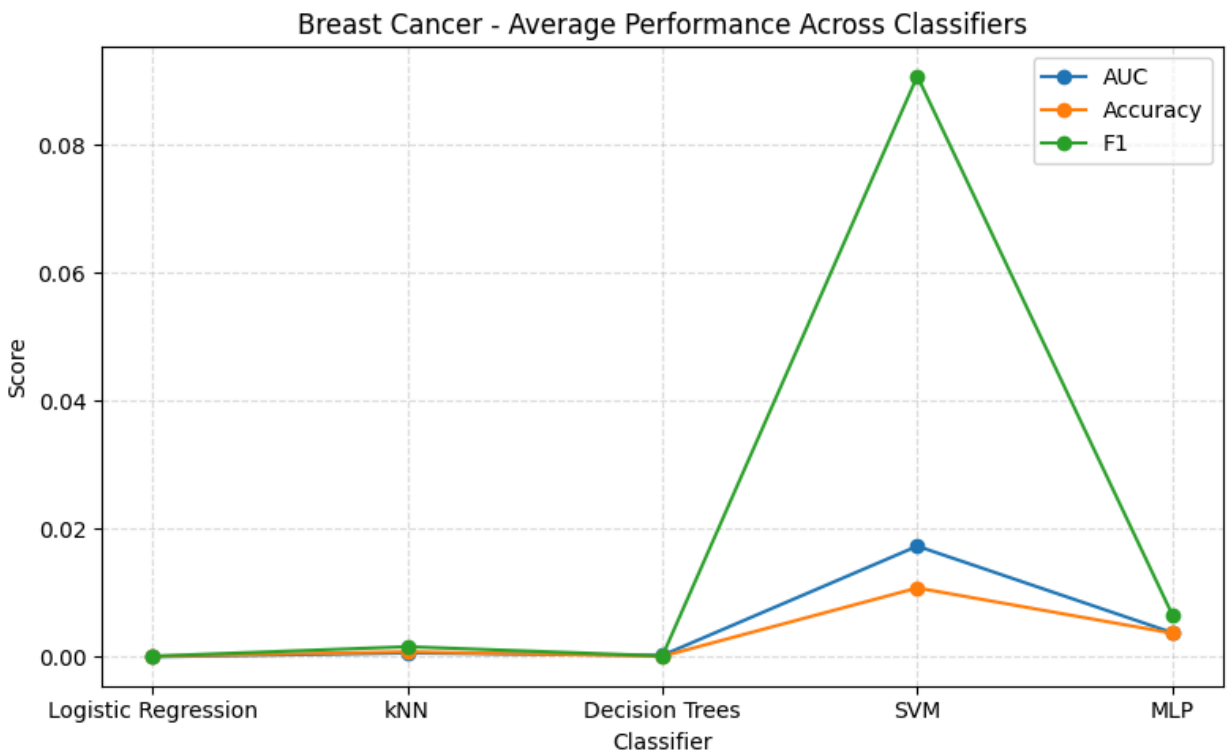
Most Stable Datasets:

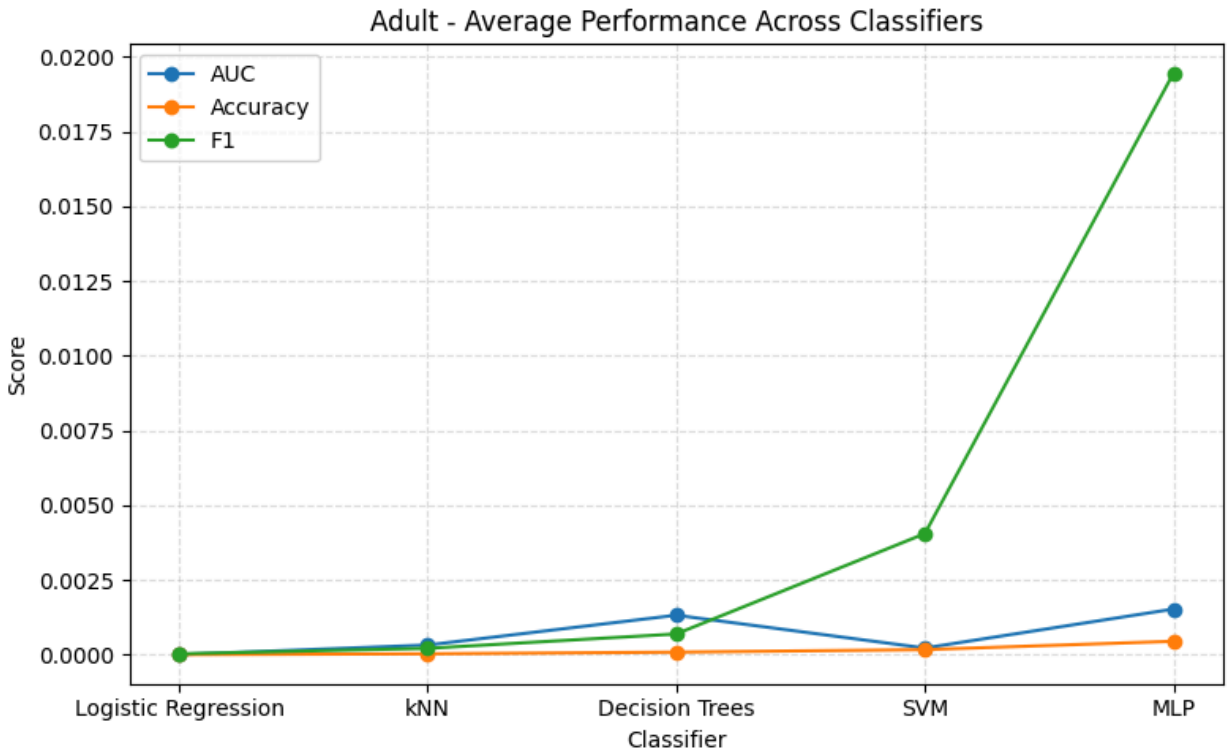
1. Iris (Low dimensionality and clear class separation)
2. Wine Quality (Large sample size smooths metric fluctuations)
3. Adult (High data volume and stable structure)




Heart - Average Performance Across Classifiers
AUC
Accuracy
F1
Score
0.04
0.03
0.02
0.01
0.00
Logistic Regression
kNN
Decision Trees
SVM
MLP
Classifier


Wine - Average Performance Across Classifiers
AUC
Accuracy
F1
Score
0.00200
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
Logistic Regression
kNN
Decision Trees
SVM
MLP
Classifier


Breast Cancer - Average Performance Across Classifiers
AUC
Accuracy
F1
Score
0.08
0.06
0.04
0.02
0.00
Logistic Regression
kNN
Decision Trees
SVM
MLP
Classifier


Adult - Average Performance Across Classifiers
AUC
Accuracy
F1
Score
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
Logistic Regression
kNN
Decision Trees
SVM
MLP
Classifier

5. CONCLUSION AND FUTURE DIRECTIONS

This study demonstrates that hyperparameter sensitivity depends heavily on both **dataset characteristics** and **classifiers**.

Key Findings

- **By Classifier:**
  - **Neural networks** (MLP) and **kernel-based** methods (SVM) show high sensitivity to hyperparameter changes, especially those controlling learning dynamics or regularization.
  - In contrast, **distance-based** (KNN), **linear** (Logistic Regression), and **tree-based** models (Decision Trees) generally demonstrate greater stability across most hyperparameter ranges.
- **By Dataset:**
  - **Smaller datasets with complex class boundaries** (e.g., Heart Disease, Breast Cancer) are more prone to sensitivity.
  - **Larger or linearly separable datasets** (e.g., Wine Quality, Adult) exhibit more stable metric behavior across tuning.
- **By Hyperparameter:**
  - **Regularization hyperparameters** (e.g., C in SVM and Logistic Regression, alpha in MLP) are consistently the most sensitive, as they directly control model complexity and generalization.
  - **Structural hyperparameters** such as max_depth (Decision Trees), n_neighbors (KNN), and hidden_layer_sizes (MLP) also exhibit high sensitivity by shaping the model's capacity and decision boundaries.
  - In contrast, **algorithmic choices** like solver (Logistic Regression, MLP) and activation functions (MLP) tend to have relatively lower impact, showing minimal variance in performance across their values.

Future Work

- Extend study to Random Forests, Gradient Boosting, XGBoost and other classifiers.
- Analyze interaction effects between hyperparameters, using grid search or other methods
- Explore automated tuning algorithms (Bayesian optimization) relative to sensitivity maps.